UNIT-5

ENGINEERING POLYMERS AND NANOCOMPOSITES

Abstract
Unit 5 discuss about various types of polymers and their specific application in day today life. Various structural cComposite materials used for mechanical applications and automobile applications are based on their performance.. Various Thermosets resins- bakelite and epoxy & thermoplastics- polycarbonate and polyether sulfone are discussed. Processing parameter like tg of polymers is discussed along with biodegradable polymers which are the sustainable solution for the existing nondegradable polymers. Various testing methods are discussed to get an insight about the performance evaluation of the composite materials.

Blow-up Syllabus

Sl No	Topic to be taught	Duration
1	Tg-Significance of Tg and factors affecting Tg-	1 hour
2	Thermosets-bakelite and epoxy- preparation and specific applications in industries	1 hour
3	thermoplastics- polycarbonate and polyether sulfones- preparation and specific applications in industries.	1 hour
4	Biodegradable polymer: Introduction, synthesis, properties, and application of poly lactic acid (PLA).	1 hour
5	Glass fibre-synthesis, properties and applications in polymer composites.	1 hour
6	carbon and natural fibre - synthesis, properties and applications in polymer composites	1 hour
7	ASTM standards of material testing-tensile strength, flexural strength,	1 hour
8	ASTM standards of material testing- ILSS and impact strength.	1 hour
9	Applications of polymer nanocomposites in injection moulded products, paints and 3D printing.	1 hour

UNIT-5 ENGINEERING POLYMERS & NANOCOMPOSITES

Thermosets-Bakelite and Epoxy, thermoplastics- polycarbonate and polyether sulfones-preparation and specific applications in industries. Biodegradable polymer: Introduction, synthesis, properties, and application of poly lactic acid (PLA). Significance of glass transition temperature (Tg) and factors affecting Tg.

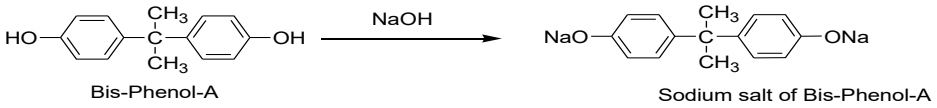
Reinforcements and testing- Glass, carbon and natural fibre - synthesis, properties and applications in polymer composites. ASTM standards of material testing-tensile strength, flexural strength, ILSS and impact strength. Applications of polymer nanocomposites in injection moulded products, paints and 3D printing.

THERMOSETTING POLYMERS:

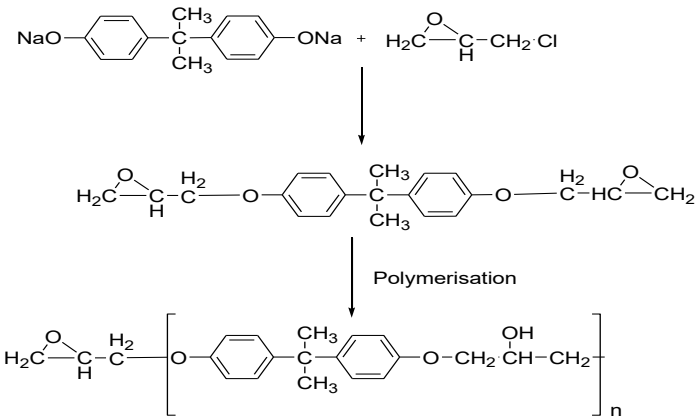
Some polymers undergo chemical change on heating and convert themselves into an infusible mass. They are like the yolk of egg, which on heating sets into a mass, and, once set, cannot be reshaped. Such polymers, that become infusible and insoluble mass on heating, are called "thermosetting" polymers. These polymers are cross linked or heavily branched molecules, which on heating undergo extensive cross linking in moulds and again become infusible. These cannot be reused. Some common examples are epoxy resins, bakelite, urea-formaldelyde resins, etc.

Epoxy resins: (Araldite)

These are the polymers contain epoxy group on both the ends. The most common epoxy resin is obtained by condensation reaction of epichlorohydrin and Bis-phenol-A in the presence of NaOH as catalyst.



Bis-Phenol-A Sodium salt of Bis-Phenol-A



Properties:

1. Epoxy resins possess good electrical insulating properties.
2. Epoxy resins are resistant to water, acids, alkalis and various organic solvents.
3. They offer very good wearing and abrasion resistance.

4. The polar nature of the resins makes them excellent adhesives.

5. Cross linked polymers have high toughness and heat resistant properties.

Applications:

1. It is used as an adhesive to bind wood, glass, concrete, ceramic, metallic and leather materials.

2. Epoxy resins are applied over cotton, rayon and bleached fabrics to impart grease resistance and shrinkage control.

3. Used in manufacture of skid resistant industrial floorings and highway surfacing.

4. Used as laminating materials for electrical equipments.

5. Moulds which are made from epoxy resins are used in the production of aircraft and automobile components.

Note: Curing of epoxy resins

Epoxy resins are converted in to three dimensional cross linked polymers by reaction with curing agents such as polyamides, polyamines, dicarboxylic acids, acid anhydrides, etc. The process is called curing of epoxy resins. On curing epoxy resins show outstanding properties of toughfness, chemical inertness, flexibility and strong adhesion.

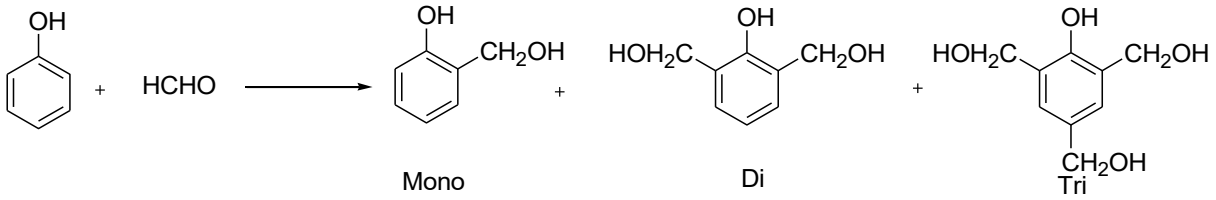
Phenol Formaldehyde resins:

Phenol-Formaldehede (PF) resins are formed by the poly condensation between phenol and formaldehyde. The polycondensation reaction can be catalysed either acids or bases.

Synthesis:

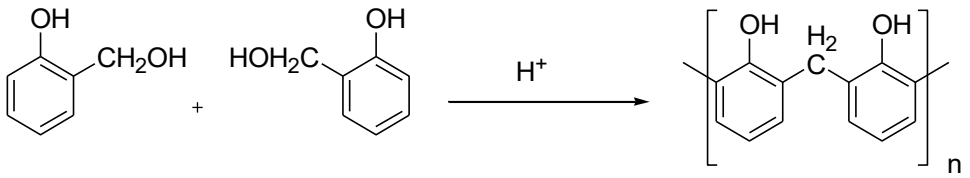
Aqueous solution of formaldehyde with phenol is heated to 70-75 C. The monomers like monomethylol phenol, dimethylol phenol, trimethylol phenol are formed. These monomers condense to give a polymer. The nature of the reaction depends on the ratio of phenol to formaldehyde in the solution.

Phenol reacts with formaldehyde forms methylol phenols as below.



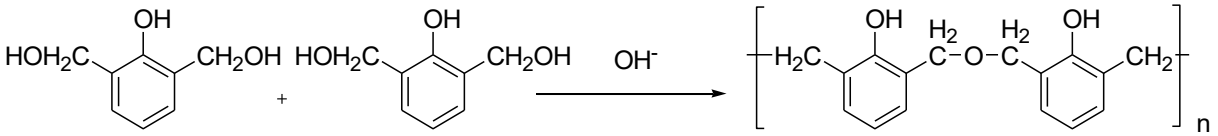
The type of methylol phenol formed will depend on the ratio of phenol to formaldehyde in solution.

Case (i) If the phenol to formaldehyde ratio is is greater than one (P/F>1), the monomethylol phenol is expected in higher percentage. This monomethylol phenol undergoes polymerisation in presence of acidic catalyst forms a polymer called Novolac.



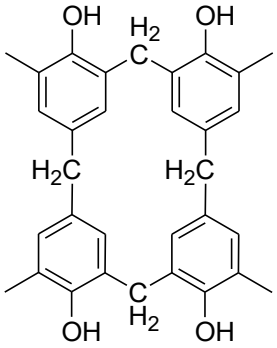
Novolac

Case (ii) If the phenol to formaldehyde ratio is less than one (P/F<1), the di and trimethylol phenol are expected in higher percentage. This di and trimethylol phenols undergo polymerisation in presence of base catalyst forms a polymer called Resol.

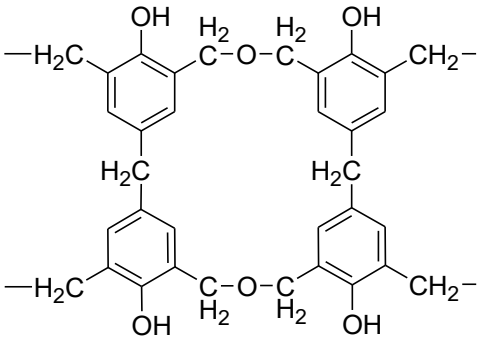


Resol

Novolac or resol undergo curing when mixed with hexamine (Hexamethylene tetramine). Hexamine helps in the cross linking of the chains. The highly cross linked product is called Bakelite.



Bakelite from Novolac



Bakelite from Resol

Note:

1. Novolac or resol resins when mixed with drying oil are used as varnishes.
2. Both novolac and resol are linear polymers therefore thermoplastic in nature.
3. Resol is characterised by ether linkage where as novolac is characterised by methylene linkage.
4. The curing agent, hexamine acts as a source of formaldehyde when it reacts with water.

Properties of phenol-Formaldehyde resins:

1. Bakelite is resistant to water, acids, and organic solvents.
2. Due to the presence of phenolic –OH group it is attacked by alkalies.
3. Good electrical insulator and got excellent adhesive properties.
4. Rigid, hard and scratch resistant thermosetting polymer.

Applications:

1. Used in the manufacture of heater handles, TV and radio cabinets, because of it high temperature resistance.
2. As Bakelite is good insulator, it is used for making electric insulator parts like switches, plugs, and switch boards.
3. For making brake linings, abrasive wheels and sand papers.
4. To seal electrical bulbs to their caps.
5. For making decorative laminates and wall papers.
6. In the production of ion exchange resins used in the purification of water.

THERMO PLASTIC POLYMERS:

Some polymers soften on heating and can be converted into any shape that they can retain on cooling. The process of heating, reshaping and retaining the same on cooling can be repeated several times. Such polymers, that soften on heating and stiffen on cooling, are termed 'thermoplastics'. These are the linear or slightly branched long chain molecules capable of repeatedly softening on heating and hardening on cooling. These polymers possess intermolecular forces of attraction intermediate between elastomers and fibres. Polyethylene, PVC, nylon and sealing wax are examples of thermoplastic polymers.

Poly carbonate:

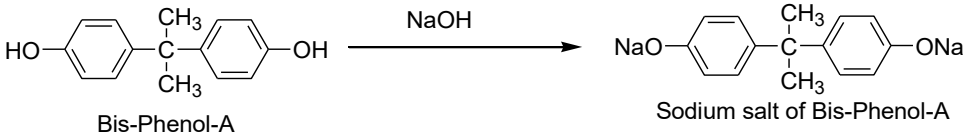
Polycarbonates are the polyesters of unstable carboxylic acids. They contain [-O-CO-O-] linkage through the chain.

Polycarbonates are prepared by condensation of

(i) Posgene and sodium salt of Bis-phenol-A

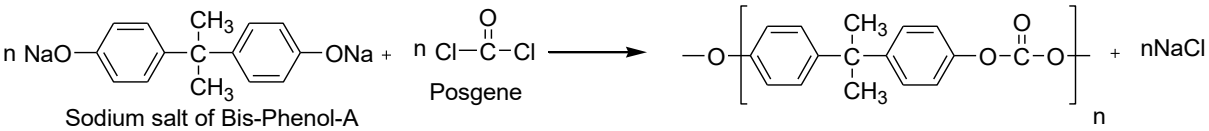
(ii) Diphenyl carbonate and Bis-phenol-A

Method(i): From Posgene and sodium salt of Bis-phenol-A



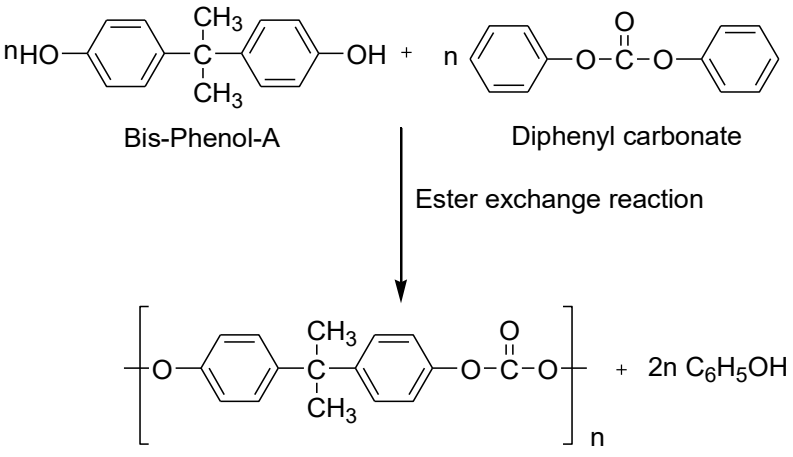
Bis-Phenol-A

Sodium salt of Bis-Phenol-A



Sodium salt of Bis-Phenol-A

Method (ii): From Diphenyl carbonate and Bis-phenol-A



Properties:

1. It has high tensile strength and impact resistance.

2. It is white transparent thermoplastic polymer.

3. Highly soluble in alkalies and acids

4. Got very high melting point.

Applications:

1. Used in the manufacture of many useful articles such as telephone parts, unbreackable glazing appliences, etc due to its high tensile strength.

2. Used in the manufacture of safety goggles, automobile lenses, DVD, CD etc due to its transparent nature.

POLY ETHER SULFONES

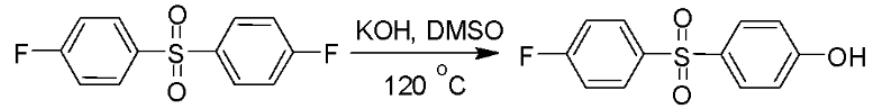
Polyethersulfones are one of the most important high performance thermo plastic polymers. These polymers are well known for their high toughness, temperature and chemical stability. Generally,

these polymers contain hydrophilic sulfone functional groups along with hydrophobic aromatic moieties in their back bone. These polymers can withstand high temperature ranges from 100-200 oC, due to this property they can be used in electrical and automobile applications.

Synthesis

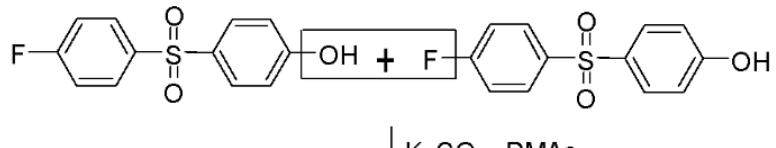
These polymers can be synthesized from bisfluorophenyl sulfones. In the first step bis fluorosulfones are made to react with KOH to convert one of its fluoro functional to -OH in highly protic solvent Dimethyl sulfoxide (DMSO). Then this monomer is polymer is polymerized using condensation polymerization technique in presence of Dimethylacetamide at very high temperature 160°C using K_2CO_3 as base

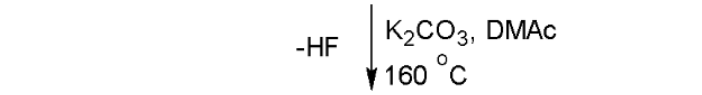
Step 1

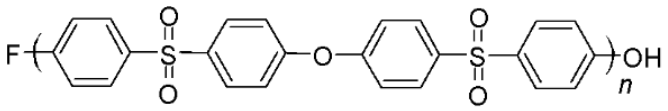


1,1'-sulfonylbis(4-fluorobenzene) 4-(4-fluorobenzene-1-sulfonyl)phenol

Step 2







Poly ether sulfones

Properties

Due to rigid structure of poly sulfone, they are very stable and exhibit excellent properties such as

- Excellent heat resistance
- Hydrolysis resistance to aqueous and alkaline media
- Good electrical resistance
- High mechanical strength with excellent young's and rigidity modulus
- Highly transparent
- Excellent flame retardancy
- The glass transition temperature ranges from 190-230 oC
- and transparent
- High dimensional stability
- Highly resistant to mineral acids, alkali, and electrolytes, in various pH
- It is also stable in aqueous acids and bases and many non-polar solvents

Applications

- In the preparation of membrane for water purifications
- For gas and chemical separations using membrane technology
- Used as composite materials along with glass fibres in mechanical applications
- As membrane in proton exchange membrane fuel cells
- As food storage containers because of their thermal stability
- In electrical equipment's
- Because of their thermal stability it is extensively used in aerospace applications
- As heat retardant materials in automobiles

BIODEGRADABLE POLYMERS:

It is observed that plastic bucket kept in sunlight and rain for long time loses its lustre and strength. This deterioration in properties is due to a phenomenon called „polymer degradation", which is characterised by an uncontrolled change in the molecular weight or constitution of the polymer. Conventionally, the degradation is a

reduction in the molecular weight of the polymer.

In developing countries, environmental pollution by synthetic polymers has assumed dangerous proportions. Petroleum-derived plastics are not readily biodegradable and because of their resistance to microbial degradation, they accumulate in the environment. In addition in recent times oil prices have increased markedly. These facts have helped to stimulate interest in biodegradable polymers. Biodegradable plastics and polymers were first introduced in 1980s. Polymers from renewable resources have attracted an increasing amount of attention over the last two decades, predominantly due to two major reasons: firstly environmental concerns, and secondly the realization that our petroleum resources are finite. There are many sources of biodegradable plastics, from synthetic to natural polymers. Natural polymers are available in large quantities from renewable sources, while synthetic polymers are produced from non-renewable petroleum resources. Biodegradation of polymeric biomaterials involves cleavage of hydrolytically or enzymatically sensitive bonds in the polymer leading to polymer erosion. A vast number of biodegradable polymers have been synthesized recently and some microorganisms and enzymes capable of degrading them have been identified.

"Thus the biodegradable polymers are the polymers which will degrade by the action of naturally occurring microorganisms like bacteria, fungi or sunlight".

Poly Lactic acid (PLA):

The basic building block of PLA is the lactic acid (LA). It is a simple chiral molecule which exists as two enantiomers, L- and D-lactic acid, optically active. It is can be produced by fermentative or chemical synthesis. Today the most popular route is fermentation, in which sugars and starches are converted into lactic acid by bacterial fermentation using an optimized strain of Lactobacillus.

Properties: The PLA is a semi-crystalline polymer with glass transition temperature around 55 to 59°C and melting point 174-184 ° C. It shows a good mechanical strength, high Young's modulus, thermal plasticity and has good processability. It is unstable in wet conditions, which can undergo chain disruption in the human body and degrades into nontoxic by products, lactic acid, carbon dioxide and water which are subsequently eliminated through the Krebs cycle and in the urine.

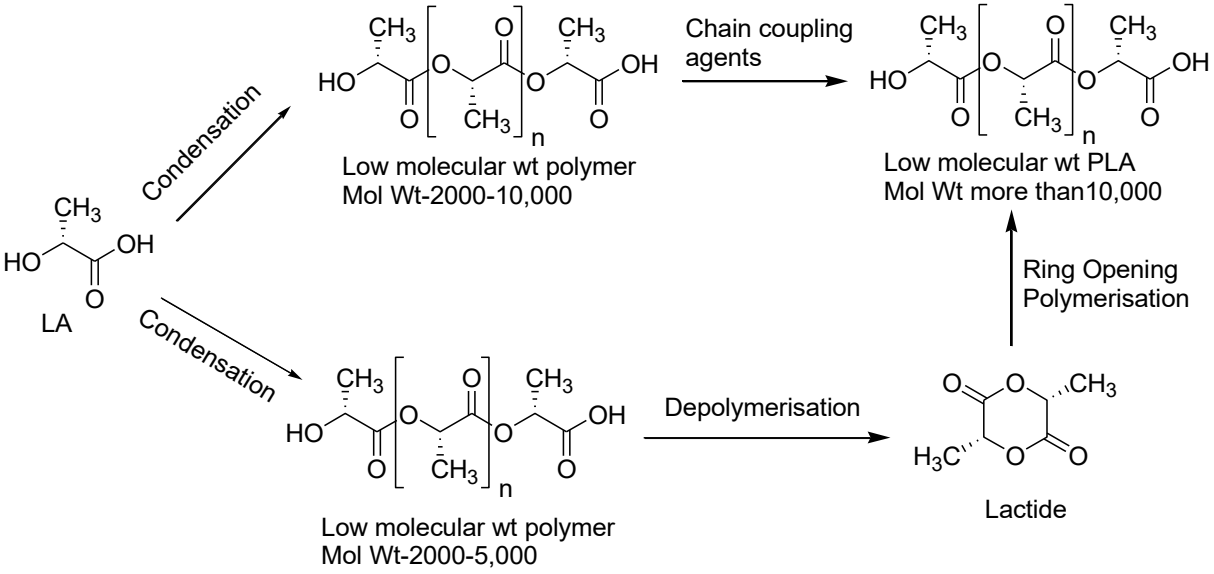
Synthesis of Poly Lactic Acid:

There are two important methods for PLA synthesis:

(i) Direct polycondensation of lactic acid and

(ii) Ring opening polymerization of lactic acid cyclic dimmer, known as lactide.

Following reactions show the reaction mechanism for both of them.



In direct condensation, solvent is used and higher reaction times are required. The resulting polymer is a material of low to intermediate molecular weight.

Ring-opening polymerization (ROP) of the lactide needs catalyst (Stannous octoate ($Sn(Oct)_2$)) but results in PLA with controlled molecular weight. Depending on monomer used and controlling reactions conditions, it is possible to control the ratio and sequence of Dand L-lactic acid units in the final polymer.

Applications of biodegradable polymers:

Medical Applications

- Wound management
  - Sutures
  - Staples
  - Clips
  - Adhesives
  - Surgical meshes

- Orthopedic devices
  - Pins
  - Rods
  - Screws
  - Tacks
  - Ligaments

PLA are used in Fracture fixation and Ligament augmentation

GLASS TRANSITION TEMPERATURE (Tg)

In crystalline solids (Eg:KCl) there will be orderly arrangement of particles, where as amorphous solids particles are arranged randomly. In-case of crystalline solids a definite quantity of energy is required to separate the particles. Hence crystalline solids show sharp melting points. But in case of amorphous solids melting points are not sharp.

Polymers are generally amorphous in nature as the polymer chains are coiled, interwined and entangled giving a random arrangement. Therefore behaviour of polymers on heating is different from that of crystalline solids.

A solid polymer will be hard, stiff and glossy as the chains occupy definite positions in the solid and virtually all the chain movements are frozen. When temperature increases the localised units (chain segments) within a long chain molecule starts moving. This movement is called segmental motion. If the temperature increases further the whole chain starts moving. This type of movement is called molecular motion.

Eg: The entangled noodles whose ends can be moved but not the whole strand, is segmental motion. Earth worm in a bowl that keeps wagging its head and tail without actually moving, is segmental motion.

The parts of polymer where the local segments have started moving is like liquid phase where as the molecule as a whole whose mobility is forbidden is like solid phase. This state is a combination of solid and liquid state and it is called rubbery state.

The temperature at which a polymer transforms from stiff, hard, glossy state to rubbery state is called glass transition temperature (Tg). The temperature at which the polymer melts and starts flowing is called flow temperature. (T_f or T_m).

SIGNIFICANCE OF Tg

i) It is used as a measure for evaluating the flexibility of a polymer and the type of response the polymeric material would exhibit to mechanical stress.

ii) Tg is very useful in choosing the right processing temperature for fabrication (molding, calendering and extrusion)

iii) Tg is very useful in determining the coefficient of thermal expansion, heat resistant refractive index, electrical property etc.

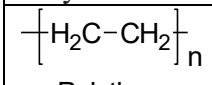
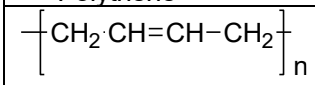
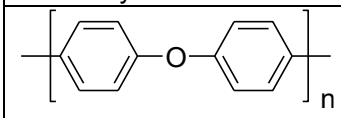
FACTORS AFFECTING GLASS TRANSITION TEMPERATURE:

Any structural features or externally imposed conditions that influence chain mobility will also affect the value of Tg. Some of these structural factors include chain flexibility; stiffness, including steric hindrance, polarity, or interchain attractive forces; geometric factors;

copolymerization; molecular weight, branching; cross-linking; and crystallinity. External variables are plasticization, pressure, and rate of testing.

1.Chain Flexibility: Chain flexibility is determined by the ease with which rotation occurs about primary valence bonds. If there exists a free rotation along carbon chain of the polymer the polymer is flexible. Linear polymer chains containing only single bonds have high degree of freedom for rotation and are more flexible. More the flexibility more will be the segmental mobility and hence lower will be the Tg

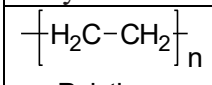
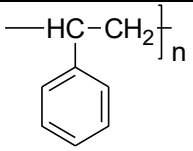
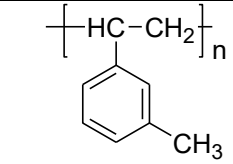
Eg:

Polymer	Tg	Remarks
 Polythene	-110° C	Free rotation is possible
 Polybutadiene	-85° C	Contains double bond so there will be restricted rotation
 Polypheneleneoxide	+83° C	Free rotation restricted, not flexible (Rigid)

2. Presence of side groups: (Bulky groups)

Presence of side groups restricts the free rotation or movement of polymer chain the polymer becomes more rigid and the Tg increases.

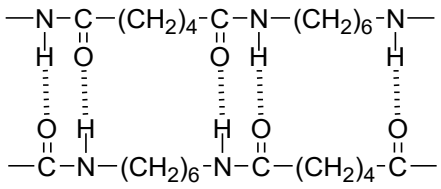
Eg:

Polymer	Tg	Remarks
 Polythene	-110° C	No side groups, free rotation is possible
 Polystyrene	100° C	Presence of phenyl groups on the polymer backbone, becomes rigid, so higher Tg
 Methyl substituted Polystyrene	170° C	Presence of methyl substituted phenyl groups on the polymer backbone, becomes still rigid, so still higher Tg

3. Inter molecular forces:

Presence of intermolecular forces in the polymer chain due to polar groups, dipoles etc exerts strong force of attraction like hydrogen bonding with neighbouring chains. This restricts the segmental mobility and hence increases the Tg.

Eg: Tg of polypropylene is -18°C where as taht of nylon-6,6 is 57°C. This is because nylon-6,6 contains polar amide bonds, there will be hydrogen bonding as shown below.



4. Molecular mass:

Tg increases with increase in the molecular mass up to 20,000. Molecular mass beyond 20,000 will have negligible effect on Tg. Under high molecular mass, long polymeric chains coil and entangle with one another. This restricts free mobility of the chains and thus increases Tg. Under low molecular mass, more number of small chains will have number of loose ends. This leads to segmental motion at the loose end resulting in lower Tg.

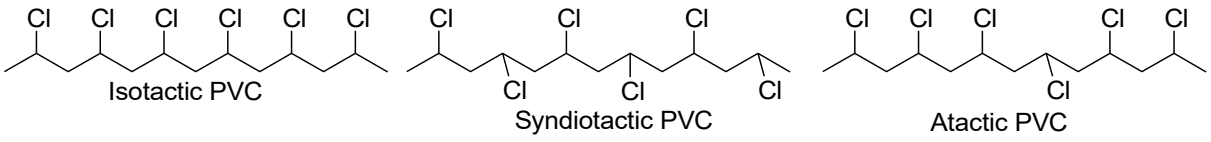
5. Crystallinity:

Close and orderly arrangement of chains leads to crystallinity in a polymer. When chains are close (as in crystalline solid) intermolecular force of attraction (Vaander Walls force) will be more. So more energy is required to initiate segmental motion which leads to high Tg. (Higher crystallinity is expected if the chains are linear without branching).

6. Stereoregularity of the polymer:

Isotactic polymer is more symmetric than syndiotactic, which is more symmetric than atactic polymer. When the structure is symmetric, chains can come closer which leads to increase in crystallinity and thereby increasing Tg of the polymer. Thus, Tg of these polymers is in the order of**syndiotactic > isotactic>atactic polymer.**

Eg: Three different confirmations of PVC are as follows.



REINFORCEMENTS & TESTING

Reinforcement

The role of the reinforcement in a composite material is fundamentally one of increasing the mechanical properties of the neat resin system. All of the different fibres used in composites have different properties and so affect the properties of the composite in different ways. The properties and characteristics of common fibres are explained below.

However, individual fibres or fibre bundles can only be used on their own in a few processes such as filament winding (described later). For most other applications, the fibres need to be arranged into some form of sheet, known as a fabric, to make handling possible. Different ways for assembling fibres into sheets and the variety of fibre orientations possible lead to there being many different types of fabrics, each of which has its own characteristics.

Fibre Properties

The mechanical properties of most reinforcing fibres are considerably higher than those of un-reinforced resin systems. The mechanical properties of the fibre/resin composite are therefore dominated by the contribution of the fibre to the composite.



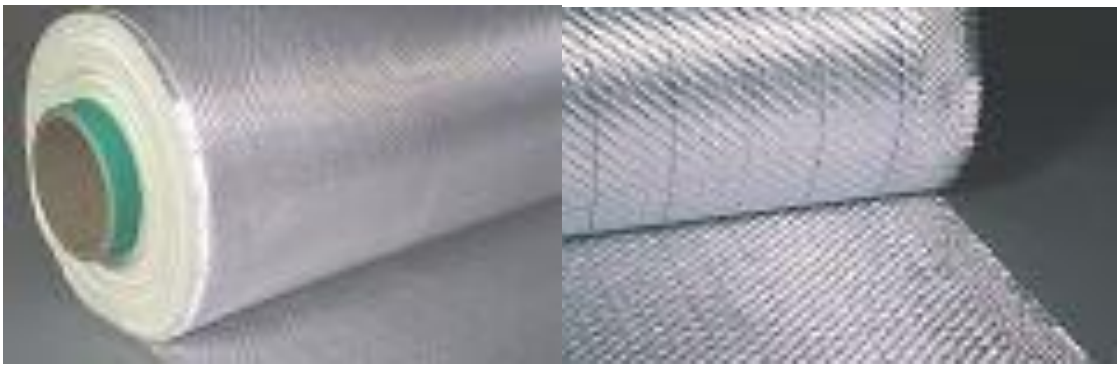
The four main factors that govern the fibre's contribution are:

1.The basic mechanical properties of the fibre itself.
2.The surface interaction of fibre and resin (the 'interface').
3.The amount of fibre in the composite ('Fibre Volume Fraction').
4. The orientation of the fibres in the composite.

The basic mechanical properties of the most commonly used fibres are later. The surface interaction of fibre and resin is controlled by the degree of bonding that exists between the two. This is heavily influenced by the treatment given to the fibre surface, and a description of the different surface treatments and 'finishes' is also given here.

The amount of fibre in the composite is largely governed by the manufacturing process used. However, reinforcing fabrics with closely packed fibres will give higher Fibre Volume Fractions (FVF) in a laminate than will those fabrics which are made with coarser fibres, or which have large gaps between the fibre bundles. Fibre diameter is an important factor here with the more expensive smaller diameter fibres providing higher fibre surface areas, spreading the fibre/matrix interfacial loads. As a general rule, the stiffness and strength of a laminate will increase in proportion to the amount of fibre present. However, above about 60-70% FVF (depending on the way in which the fibres pack together) although tensile stiffness may continue to increase, the laminate's strength will reach a peak and then begin to decrease due to the lack of sufficient resin to hold the fibres together properly.

Finally, since reinforcing fibres are designed to be loaded along their length, and not across their width, the orientation of the fibres creates highly 'direction-specific' properties in the composite. This 'anisotropic' feature of composites can be used to good advantage in designs, with the majority of fibres being placed along the orientation of the main load paths. This minimises the amount of parasitic material that is put in orientations where there is little or no load



GLASS FIBER-REINFORCED POLYMER (GFRP) COMPOSITES

Fiberglass is simply a composite consisting of glass fibers, either continuous or discontinuous, contained within a polymer matrix; this type of composite is produced in the largest quantities. The composition of the glass that is most commonly drawn into fibers (sometimes referred to as E-glass) is contained in Table 13.1; fiber diameters normally range between 3 and 20 m. Glass is popular as a fiber reinforcement material for several reasons:

1. It is easily drawn into high-strength fibers from the molten state.

2. It is readily available and may be fabricated into a glass-reinforced plastic economically using a wide variety of composite-manufacturing techniques.

3. As a fiber it is relatively strong, and when embedded in a plastic matrix, it produces a composite having a very high specific strength.

4. When coupled with the various plastics, it possesses a chemical inertness that renders the composite useful in a variety of corrosive environments.

The surface characteristics of glass fibers are extremely important because even minute surface flaws can deleteriously affect the tensile properties. Surface flaws are easily introduced by rubbing or abrading the surface with another hard material. Also, glass surfaces that have been exposed to the normal atmosphere for even short time periods generally have a weakened surface layer that interferes with bonding to the matrix. Newly drawn fibers are normally coated during drawing with a "size," a thin layer of a substance that protects the fiber surface from damage and undesirable environmental interactions. This size is ordinarily removed prior to composite fabrication and replaced with a "coupling agent" or finish that produces a chemical bond between the fiber and matrix. There are several limitations to this group of materials. In spite of having high strengths, they are not very stiff and do not display the rigidity that is necessary for some applications (e.g., as structural members for airplanes and bridges). Most fiberglass materials are limited to service temperatures below at higher temperatures, most polymers begin to flow or to deteriorate. Service temperatures may be extended to approximately by using high-purity fused silica for the fibers and high-temperature polymers such as the polyimide resins. Many fiberglass applications are familiar: automotive and marine bodies, plastic pipes, storage containers, and industrial floorings. The transportation industries are utilizing increasing amounts of glass fiber-reinforced plastics in an effort to decrease vehicle weight and boost fuel efficiencies. A host of new applications are being used or currently investigated by the automotive industry.

CARBON FIBER-REINFORCED POLYMER (CFRP)

Composites Carbon is a high-performance fiber material that is the most commonly used reinforcement in advanced (i.e., nonfiberglass) polymer-matrix composites. The reasons for this are as follows:

1. Carbon fibers have the highest specific modulus and specific strength of all reinforcing fiber materials.

2. They retain their high tensile modulus and high strength at elevated temperatures; high-temperature oxidation, however, may be a problem.

3. At room temperature, carbon fibers are not affected by moisture or a wide variety of solvents, acids, and bases.

4. These fibers exhibit a diversity of physical and mechanical characteristics, allowing composites incorporating these fibers to have specific engineered properties.

5. Fiber and composite manufacturing processes have been developed that are relatively inexpensive and cost effective.

Use of the term “carbon fiber” may seem perplexing since carbon is an element, the stable form of crystalline carbon at ambient conditions is graphite. Carbon fibers are not totally crystalline, but are composed of both graphitic and noncrystalline regions; these areas of noncrystallinity are devoid of the three-dimensional ordered arrangement of hexagonal carbon networks that is characteristic of graphite. Manufacturing techniques for producing carbon fibers discussed in previous unit. However, three different organic precursor materials are used: rayon, polyacrylonitrile (PAN), and pitch. Processing technique will vary from precursor to precursor, as will also the resultant fiber characteristics. One classification scheme for carbon fibers is by tensile modulus; on this basis the four classes are standard, intermediate, high, and ultrahigh moduli. Furthermore, fiber diameters normally range between 4 and 10 m; both continuous and chopped forms are available. In addition, carbon fibers are normally coated with a protective epoxy size that also improves adhesion with the polymer matrix.

Carbon-reinforced polymer composites are currently being utilized extensively in sports and recreational equipment (fishing rods, golf clubs), filament-wound rocket motor cases, pressure vessels, and aircraft structural components—both military and commercial, fixed wing and helicopters (e.g., as wing, body, stabilizer, and rudder components).

SYNTHETIC FIBRES:
Glass fiber

Textile-grade glass fibers are made from silica (SiO_2) sand, which melts at 1720°C/3128°F. SiO_2 is also the basic element in quartz, a naturally occurring rock. Quartz, however, is crystalline (rigid, highly ordered atomic structure) and is 99 percent or more SiO_2. If SiO_2 is heated above 1200°C/2192°F then cooled ambiently, it crystallizes and becomes quartz. Glass is produced by altering the temperature and cooldown rates. If pure SiO_2 is heated to 1720°C/3128°F then cooled quickly, crystallization can be prevented and the process yields the amorphous or randomly ordered atomic structure we know as glass. Although continuously refined and improved, today's glass fiber manufacturers combine this high heat/quick cool strategy with other steps in a process that is basically the same as that developed in the 1930s, albeit on a much larger scale. This process can be broken down into five basic steps: batching, melting, fiberization, coating and drying/packaging.

Step 1: Batching
Although a viable commercial glass fiber can be made from silica alone, other ingredients are added to reduce the working temperature and impart other properties that are useful in specific applications. For example, E-glass, originally aimed at electrical applications, with a composition including SiO_2, AI_2O_3 (aluminum oxide or alumina), CaO (calcium oxide or lime) and MgO (magnesium oxide or magnesia), was developed as a more alkali-resistant alternative to the original soda lime glass. Later, boron was added via B_2O_3 (boron oxide) to increase the difference between the temperatures at which the E-glass batch melted and at which it formed a crystalline structure to prevent clogging of the nozzles used in fiberization (Step 3, below). S-glass fibers, developed for higher strength, are based on a SiO_2-AI_2O_3-MgO formulation but contain higher percentages of SiO_2 for applications in which tensile strength is the most important property.

In the initial stage of glass manufacture, therefore, these materials must be carefully weighed in exact quantities and thoroughly mixed (batched). Batching has become automated, using computerized weighing units and enclosed material transport systems. For example, in Owens Corning's plant in Taloja, India, each ingredient is transported via pneumatic conveyors to its designated multistory storage bin (silo), which is capable of holding 70 to 260 ft^3 (1.98 to 7.36m^3) of material. Directly beneath each bin is an automated weighing and feeding system,

which transfers the precise amount of each ingredient to a pneumatic blender in the batch house basement.

Step 2: Melting

From the batch house, another pneumatic conveyor sends the mixture to a high temperature (≈1400ºC/2552ºF) natural gas-fired furnace for melting. The furnace is typically divided into three sections, with channels that aid glass flow. The first section receives the batch, where melting occurs and uniformity is increased, including removal of bubbles. The molten glass then flows into the refiner, where its temperature is reduced to 1370ºC/2500ºF. The final section is the forehearth, beneath which is located a series of four to seven bushings that, in the next step, are used to extrude the molten glass into fibers. Large furnaces have several channels, each with its own forehearth. Control of oxygen flow rates are crucial because furnaces that use the latest technology burn nearly pure oxygen instead of air because it helps the natural gas fuel to burn cleaner and hotter, melting glass more efficiently. It also lowers operating costs by using less energy and reduces nitrogen oxide (NOx) emissions by 75 percent and carbon dioxide (CO2) emissions by 40 percent.

The industry takes three main approaches to glass melting:

(1) indirect melt (also called marble remelt);

(2) direct melt using larger-scale furnaces (8,000 to 100,000 metric tonnes per year); and

(3) direct melt using smaller-scale furnaces (150 to 200 metric tonnes per year), which are also called paramelters.

For indirect marble remelt, molten glass is sheared and rolled into marbles roughly 0.62 inch (15 to 16 mm) in diameter, which are cooled, packaged and then transported to a fiber manufacturing facility where they are remelted for fiberization (see “Step 3”). The marbles facilitate visual inspection of the glass for impurities, resulting in a more consistent product. The direct melt process transfers molten glass in the furnace directly to fiber-forming equipment. Because direct melting eliminates the intermediate steps and the cost of forming marbles, it has become the most widely used method.

Step 3: Fiberization

Glass fiber formation, or fiberization, involves a combination of extrusion and attenuation. In extrusion, the molten glass passes out of the forehearth through a bushing made of an erosion-resistant platinum/rhodium alloy with very fine orifices, from 200 to as many as 8,000. Bushing plates are heated electronically, and their temperature is precisely controlled to maintain a constant glass viscosity. Water jets cool the filaments as they exit the bushing at roughly 1204ºC/2200ºF. Attenuation is the process of mechanically drawing the extruded streams of

molten glass into fibrous elements called filaments, with a diameter ranging from 4 μm to 34 μm (one-tenth the diameter of a human hair). A high-speed winder catches the molten streams and, because it revolves at a circumferential speed of ~2 miles/~3 km per minute (much faster than the molten glass exits the bushings), tension is applied, drawing them into thin filaments.

The bushings are expensive, and their nozzle design is critical to fiberization. Nozzle diameter determines filament diameter, and the nozzle quantity equals the number of ends. A 4,000-nozzle bushing may be used to produce a single roving product or the process can be configured to make four rovings with 1,000 ends each. The bushing also controls the fiber yield or yards of fiber per pound of glass.

Step 4: Coating

In the final stage, a chemical coating, or size, is applied. (Although the terms binder, size and sizing often are used interchangeably in the industry, size is the correct term for the coating applied, and sizing is the process used to apply it. Size is typically added at 0.5 to 2.0 percent by weight and may include lubricants, binders and/or coupling agents. The lubricants help to protect the filaments from abrading and breaking as they are collected and wound into forming packages and, later, when they are processed by weavers or other converters into fabrics or other reinforcement forms. Coupling agents cause the fiber to have an affinity for a particular resin chemistry, improving resin wetout and strengthening the adhesive bond at the fiber-matrix interface. Some size chemistries are compatible only with polyester resin and some only with epoxy while others may be used with a variety of resins.

Step 5: Drying/packaging

Finally, the drawn, sized filaments are collected together into a bundle, forming a glass strand composed of 51 to 1,624 filaments. The strand is wound onto a drum into a forming package that resembles a spool of thread. The forming packages, still wet from water cooling and sizing, are then dried in an oven, and afterward they are ready to be palletized and shipped or further processed into chopped fiber, roving or yarn. Roving is a collection of strands with little or no twist. An assembled roving, for example, made from 10 to 15 strands wound together into a multi-end roving package, requires additional handling and processing steps. Yarn is made from one or more strands, which may be twisted to protect the integrity of the yarn during subsequent processing operations, such as weaving.

Properties of Glass Fibre

- **High tensile strength.** Glass has greater tensile strength than steel wire of the same diameter, at a lower weight.
- **Dimensional stability.** Glass fibre is not sensitive to variations in temperature and hygrometry. It has a low coefficient of linear expansion.
- **High heat resistance.** Glass fabrics retain 50% of room temperature tensile strength at 370°C, 25% at 480°C, a softening point of 845°C and a melting point of 1,135°C.
- **Good thermal conductivity.** Glass fibres are great thermal insulators because of their high ratio of surface area to weight. This property makes it highly useful in the building industry.
- **Great fire resistance.** Since glass fibre is a mineral material, it is naturally incombustible. It does not propagate or support a flame. It does not emit smoke or toxic products when exposed to heat.
- **Good chemical resistance.** Glass fibre is highly resistant to attacks by most chemicals.
- **Outstanding electrical properties.** Glass fibre has a high dielectric strength and low dielectric constant. It is a great electrical insulator even at low thicknesses.
- **Dielectric permeability.** This property of glass fibre makes it suitable for electromagnetic windows.
- **Compatibility with organic matrices.** Glass fibre can vary in size and has the ability to combine with many synthetic resins and certain mineral matrices like cement.
- **Great durability.** Glass fibre is not prone to sunlight, fungi or bacteria.
- **Non-rotting.** Glass fibre does not rot and remains unaffected by the action of rodents and insects.
- **Highly economical.** It is a cost-efficient choice compared to similar materials.

Application of Glass fibre in Composites

The glass fiber–reinforced composites are used in a variety of forms to assure the required properties of usage areas as space, aerospace, automobile, energy, marine, electronic components, and defense-related special areas.

CARBON FIBERS:

Carbon fibers are novel high-performance materials. They could be described as fibers containing at least 90 % carbon obtained by the controlled pyrolysis of appropriate fibers.

Carbon atoms are bonded together in a crystal that are aligned parallel to the axis of fibre.. This alignment gives fibre high strength to volume ratio.Several thousand carbon fibres are bundled together which are then woven into fabric.

Edison in 1879 found that carbon fibers can be used as carbon filaments in electric lamps. Since the early work of Edison, the carbon fibers have been investigated and used intensively because they generally have excellent tensile properties, low densities, high thermal and chemical stabilities in the absence of oxidizing agents, good thermal and electrical conductivities, and excellent creep resistance.

In recent years, the carbon fiber industry has been growing steadily to meet the demands arising from different applications such as aerospace (aircraft and space systems), military, turbine blades, construction, lightweight cylinders and pressure vessels, medical, automobile, and sporting goods. Furthermore, in the case of the carbon fibers, the range of their applications would depend on the type of precursors used to produce the carbon fibers.

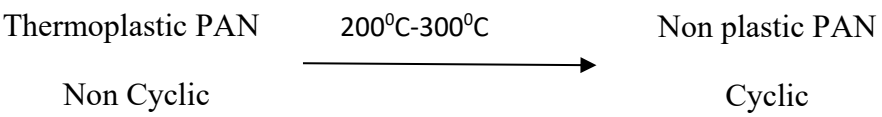
Consequently, many types of precursors have been studied to produce the carbon fibers.

Carbon fibre from Poly Acrylo Nitrile (PAN)

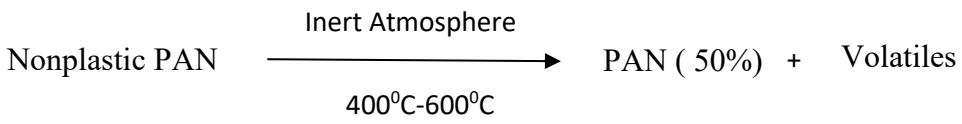
PAN-based polymers are the optimum precursors for the carbon fibers owing to a combination of tensile and compressive properties as well as the carbon yield. The PAN-based fibers were first developed by Dupont in the 1940s for use in the textile fiber. The thermal stability of PAN-based fibers was an important factor in expanding the application of fibers.

Synthesis: Synthesis of carbon fibres involves three steps.

Step-1 Oxidative stabilization- Stretching & Oxidation

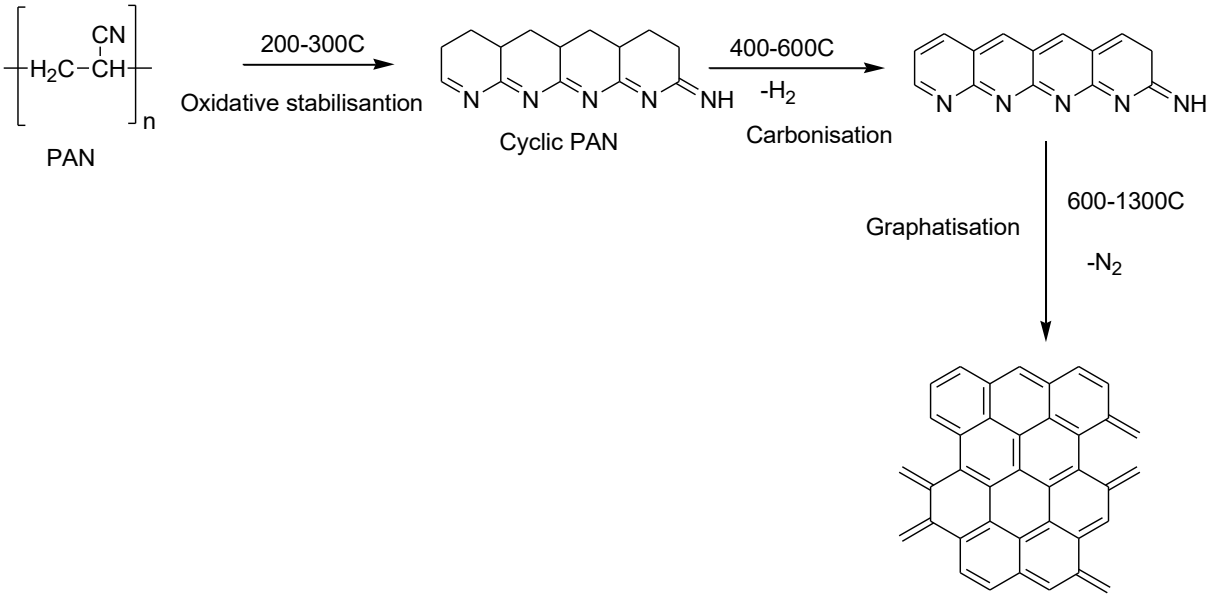


Step-II- Carbonization



Step-III-Graphatization (-N_2)

The fibres are treated at temperature between 600^{0}C- 1300^{0}C to improve order and orientation of crystallites in direction of fibre axis.



Properties of Carbon fibre

- Carbon fibre has tensile strength 3 times greater than steel & 4.5 times lighter.
- Exceptional Impact properties- Good energy dissipation mechanism
- Low coefficient of thermal expansion compared to metals
- Better corrosion resistance
- High chemical inertness

Application of carbon fibres in composites

Applications in Sports:

- Ice Hockey Sticks
- rack Spikes
- Bicycle Frames
- Helmets
- Motor Racing
- Tennis Rackets
- Golf Clubs
- Cricket Bats
- Gliders
- Surfboards
- Rowing Shells

Application in the Automotive Industry:

- Racing car chassis
- Hoods
- Car emblems
- Mufflers
- Interior panels of a car
- Steering wheels

Application in Civil Engineering

- Carbon Fiber Reinforced Polymer is can be applied to reinforce concrete structures
- The high strength of carbon fiber enables it to be used as a prestresser
- High corrosion resistance allows for use in offshore environments
- Used in PCCP (Prestressed Concrete Cylinder Pipe) lines to reinforce the pipes

Applications in Aerospace Engineering

- Aircraft: main wings, Tail units, Rudders, Elevators, Floor panel, Beams, Lavatory units, Seats
- Rockets: Nozzle cones, Motor cases
- Satellites: Antennas, Solar battery panels.owing Shells

Application in the Automotive Industry:

- Racing car chassis
- Hoods
- Car emblems
- Mufflers
- Interior panels of a car
- Steering wheels

Application in Civil Engineering

- Carbon Fiber Reinforced Polymer is can be applied to reinforce concrete structures
- The high strength of carbon fiber enables it to be used as a prestresser
- High corrosion resistance allows for use in offshore environments
- Used in PCCP (Prestressed Concrete Cylinder Pipe) lines to reinforce the pipes

Applications in Aerospace Engineering

- Aircraft: main wings, Tail units, Rudders, Elevators, Floor panel, Beams, Lavatory units, Seats
- Rockets: Nozzle cones, Motor cases
- Satellites: Antennas, Solar battery panels.

Natural Fibre

The natural fibres are usually referred to have several benefits over synthetic fibres such as availability, low cost, low density, acceptable modulus-weight ratio, high acoustic damping, low manufacturing energy consumption, low carbon footprint and biodegradable. Some authors state the evidences for clear benefits, for example, natural fibres cost much less and require much less energy to produce than traditional reinforcing fibres such as glass and carbon. However, natural fibres have negative aspects due to their low consistency of properties and quality. These fibres have higher variability of physical and mechanical properties, higher moisture absorption, lower durability, lower strength and lower processing temperature

he large variety of properties are mainly dependent upon plant species, growth conditions and method of fibre extraction. Moreover, properties depend on the fibre cell geometry of each type of cellulose and its degree of polymerization]. It should be noted that linear cellulosic macromolecules are linked by hydrogen bonds and are closely associated with hemicelluloses and lignin which confer stiffness to fibre. Not only holds fibres together but also the cellulose within the fibre cell wall

The general natural fibre life cycle phases are extraction, processing, fabrication, use, disposal and recycle. There are several limiting factors for a large-scale production and use of fibres. These factors affect several stages of the natural fibre's life cycle: large variability of soil composition and morphology, fibres hydrophilic nature, degradation by microorganisms, service life and sunlight. The physical and chemical properties of natural fibres are linked to plant source, cultivation location, climate conditions, harvest window, use of Genetic Modified Organisms (GMO), pesticides and fertilizers.

Usually, natural fibres require several treatments to overcome some of the mentioned limitations, namely to improve fibre-matrix interfacial adhesion. Several techniques have been reported, such as water-repellent chemicals, coupling agents and heat treatments, by modifying the surface morphological, topological properties, roughness and water absorption index of the fibres. As a result, research and technological effort has been reported fostering the improvement of crops quality and fibres performance on technical and economical perspective, aiming to provide new solutions and applications.

Industrial Applications of Natural Fibre Composites in the Automotive Sector

The increased importance of raw materials from renewable resources and recyclability or biodegradability of products is causing the transformation from petroleum-based synthetics to natural fibres in automotive applications. The applications of natural fibre composites in the automotive sector can be traced as far back as the 1940s, when Henry Ford produced the first composite components in a car using hemp fibre. The next referred application was in the 1950s, with the production of the body of the East German Trabant and other manufacturers also followed the same path including Daimler–Benz (1994) and Mercedes (1996). Natural fibre-based composites hold significant potential for automotive industry because the demand for light weight and environmentally friendly materials is higher. Studies indicate that natural fibre composites can contribute to a cost reduction of 20% and weight reduction of 30% of an automotive part.. Following those authors, light weight of components leads to lower fuel consumption, good recycling possibilities, reduction in waste disposal and greenhouse emissions which are some of main drivers for use of natural fibres. Natural fibre composites are mostly being used for interior parts such as dashboards, door panels, parcel shelves, seat cushions, backrests and cabin linings whereas the use of natural fibre composite parts for exterior applications is very limited.



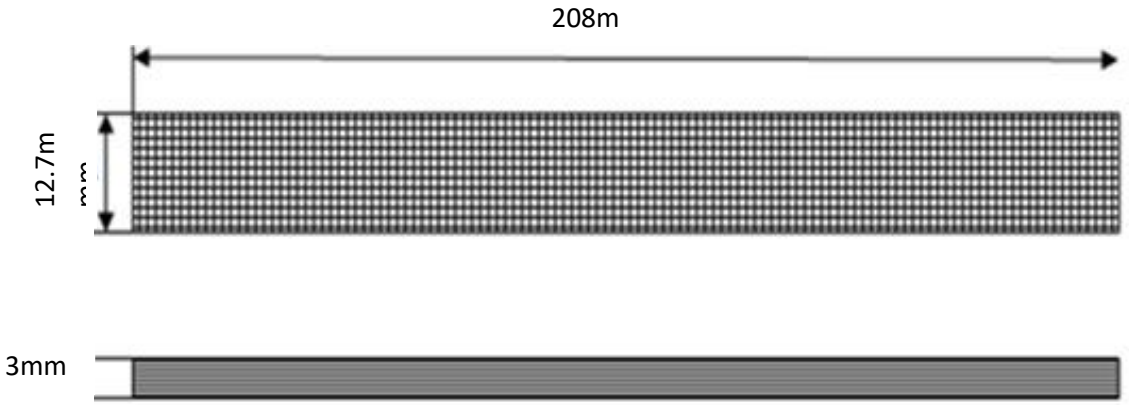
Fibres	Applications	Composition	Manufacturer
Coir	Containers, boxes, trays, packaging	Coir and natural latex rubber	Enkev
Flax	Green wall panel	50% recycled resin, reinforced with flax (25%) and E-glass (25%) roving	Innovation in green composites technology
Flax, Hemp	Racing Bicycle	Flax, Hemp and Epoxy	Museeuw Bikes
	Cases for musical instruments	Plastics (PLA and PP) and additives, natural fibres	Green line
Flax, Balsa, Wood	Summer and winter sports	Natural fibre and composite reinforcement material	Bcom
Hemp, Jute, Kenaf	Containers for shipping and storage, interior panels, load floors and underbody shields for cars and trucks, workspace panels and Furnishings for offices and homes, structural support for agricultural seedlings	Natural fibres and fiberized thermoplastic polymers	FlexForm Technologies
Kenaf	Mobile phone casing	Kenaf and PLA	NEC
Wood	Modular house construction	Wood plastic composites	Tech-wood International

MECHANICAL TESTING OF PMC'S

TENSILE TESTING OF COMPOSITES

ASTM 3039: ***Standard Test Method for Tensile Properties of Polymer Composite Materials***

The Test specimen geometry as specified in the above standard for balance symmetric glass and carbon fibre (0/90) are, width 12.7mm, length 208 mm and thickness 3 mm. The test specimen is positioned vertically in the grips of the testing machine. The grips are tightened



Tensile Test Specimen

evenly and firmly to prevent any slippage. The speed of testing employed is 5 mm/ min. Dimensions of test specimen is shown in above figure. Tensile testing is done with tension configuration grippers using a 10 ton capacity high precision computer controlled UTM. Distance between the two grippers is maintained at 50mm. The ultimate load carrying capacity of the composite laminates can be recorded.

The ultimate tensile strength (σ) is calculated using the formula

$\sigma = P/ bd \quad N/mm^2$

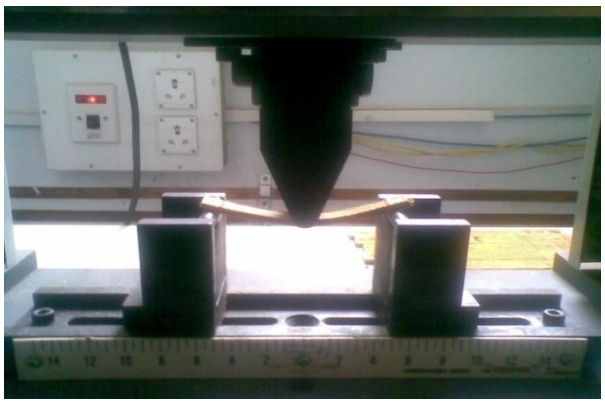
Where P = maximum load in Newton's

b = width in mm

d = thickness in mm

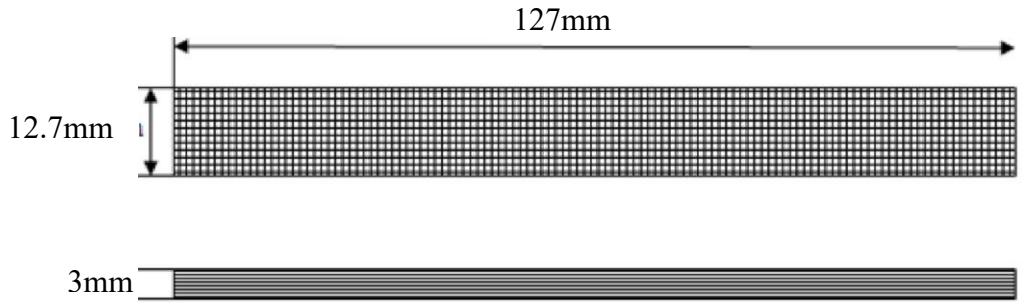
FLEXURAL TESTING OF COMPOSITES

ASTM D 790: ***Standard Test Method for Flexural Properties of Polymer Matrix Composite Laminates***



Test set up for three point bending tests

The samples are cut to the dimensions as per ASTM standards for flexural testing. The test specimen geometry as specified in the above standard for balanced symmetric carbon and glass fibre composites (0/90) are , width 12.7 mm, length 127 mm, thickness 6 mm. The test should be conducted at definite strain rate. The Test Setup is shown in figure and the details of test specimen are given in Fig shown below. Flexural test is done using a three point bend setup. The distance between the two supports are maintained at 100mm. The ultimate load carrying capacity of the composite laminates is recorded.



Flexural test specimen

(a) The flexural strength of the composites is determined using the formula

$$\sigma_f = \frac{3PL}{2bd^2}$$

Where, P = peak load in N,

L = span length in mm,

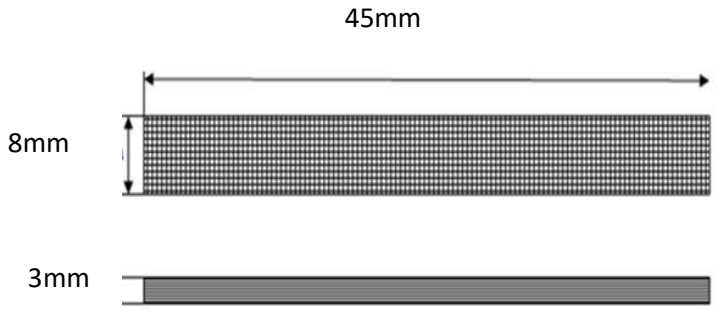
b = width of specimen in mm,

d = thickness of specimen in mm,

INTER LAMINAR SHEAR STRENGTH (ILSS) TESTING OF COMPOSITES

ASTM D 2344: ***Standard Test Method for ILSS Properties of Polymer Matrix Composite Laminates***

The delamination resistance of a material is influenced by a number of factors such as moisture, temperature, ply orientation and even crack length. The increased toughness associated with such increases in resistance (in composites) to crack propagation is traditionally associated with plastic deformation, whereas in laminates the deformations are wholly elastic. The prevalent view is that the increases come from fibres bridging, the delamination and are more extensive, though less stable, in cross-ply laminates. Three-point bend test was carried out to determine the ILSS of the specimens as per ASTM D2344.



ILSS test specimen

The testing is done on short rectangular beams of 45 x 6.35 x 3 mm at a crosshead speed of 1mm per minute. The ILSS is determined according to the following eq.

$$ILSS = \frac{0.75P}{bt} \quad \text{---------- (3.14)}$$

Where,

ILSS = interlaminar shear strength in MPa,

P = peak load in N,

b = width in mm,

t = the thickness of the specimen in mm.

IMPACT STRENGTH TESTING OF COMPOSITES

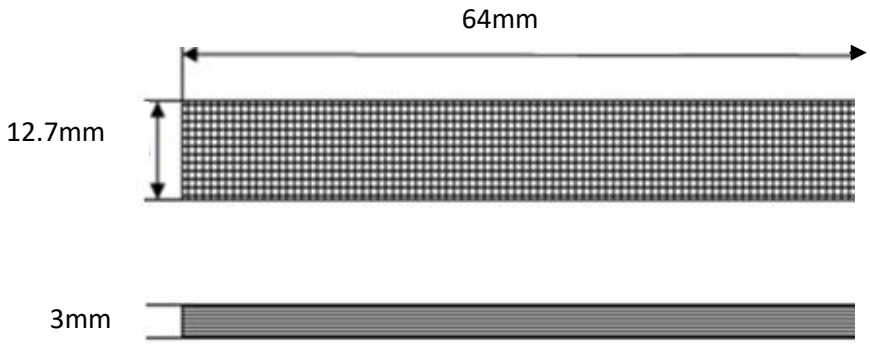
ASTM D 256: Standard test method for impact properties of polymer matrix composites

The Izod impact strength of composite samples are evaluated as per ASTM D256, using Impact Testing Machine. The test specimen geometry as specified in the above standard for balance symmetric glass and carbon fibre (0/90) composites are 64 mm long X 12.7 mm wide X 3 mm thick. The Izod test specimens should be clamped in an upright position so that the end of the specimen faced its striking edge and impact energy absorbed for breaking the specimen are obtained directly.



Izod impact testing machine

Impact strength = Impact energy in joules / thickness (mm)



Impact test specimen

This test method is designed to yield tensile fatigue data for material specifications, research and development, quality assurance, and structural design and analysis. The primary test result is the fatigue life of the test specimen under a specific loading and environmental condition. Replicate tests may be used to obtain a distribution of fatigue life for specific material types, laminate stacking sequences, environments, and loading conditions

This test method determines the fatigue behaviour of polymer matrix composite materials subjected to tensile cyclic loading. The composite material forms are limited to continuous-fiber or discontinuous-fiber reinforced composites for which the elastic properties are specially orthotropic with respect to the test direction.

Application of polymer nanocomposites in Injection moulded products, Paints & 3D Printing

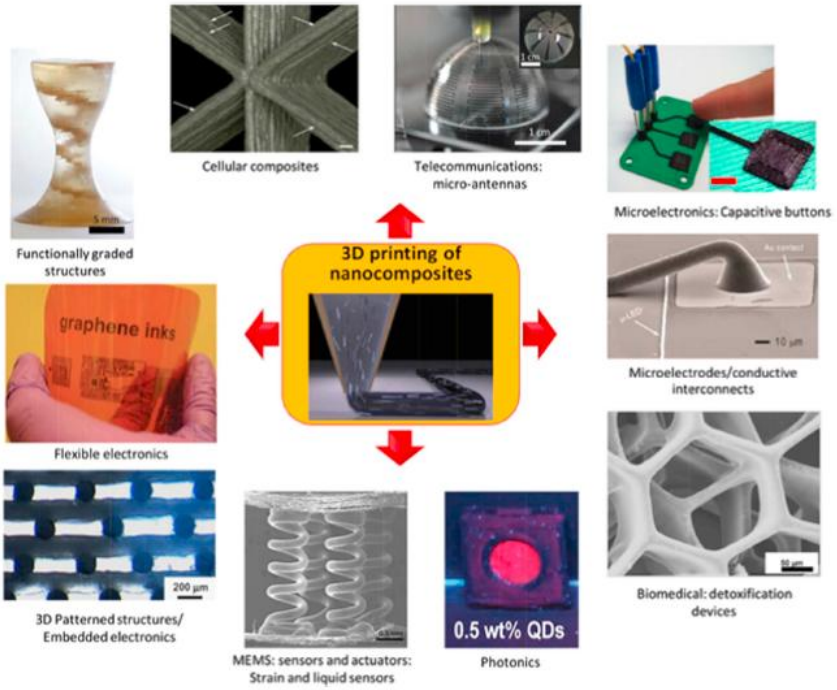
Indoor Paint

The gloss enamels are particularly used in indoor coatings as they can dry to result in highly reflecting surface. The poor fire resistance of the polymers including epoxy forms the major drawback in their practical utility. In this vein, the clay- or C-based nanomaterial–polymer nanocomposites possess improved gloss and address the above problem, thereby resulting in flame-retardant coating materials. These nanomaterials work as flame retardants by the formation of char with either carbonaceous or multilayered silicate structure in the condensed phase upon burning, which forms a protective coating onto the film surface and insulates the material coated underneath. The non-evolution of toxic gases unlike other flame-retardant additives justifies their contribution toward sustainability.

Outdoor Paint

The prime concern in the use of paints for outdoor applications is to obtain a protective finish and to guard against the deleterious effect of solar radiation, rainfall, dust, moisture, temperature, and many other environmental factors. The waterborne paints are generally recommended for use in external purposes where heavy wear and tear occurs. The nanomaterials possess resistance to corrosion, solvent, or any external factors and thus enhance the durability of the polymeric nanocomposite coatings. In this regard, the modification of clay with or functionalization of the C-based nanomaterials with benzotriazole or triazine derivatives prior to incorporation into the polymeric matrix seems promising technique to impart photostability to the paints. The flat (having 4–5 gloss at 60_ angle of incidence) or semigloss (having 20–60 or 80 gloss at 60_ angle of incidence) house paints or enamels are used for outdoor applications as they dry to a non-glossy mat surface. The polyamide modified alkyd resins find their utility as architectural and maintenance paints.

3D Printing


Functionally graded structures
Cellular composites
Telecommunications: micro-antennas
Microelectronics: Capacitive buttons
3D printing of nanocomposites
graphene inks
Flexible electronics
Microelectrodes/conductive interconnects
200 μm
3D Patterned structures/ Embedded electronics
MEMS: sensors and actuators: Strain and liquid sensors
0.5 wt% QDs
Photonics
Biomedical: detoxification devices